A. In a variational autoencoder, after training the learnt latent space distribution is characterized by mean vector μ_z = [0 0] and covariance matrix

Σ_z = [2 0
0 3]

Assuming after training, $p_\varphi(z|x) = p_\theta(x|z)$, what is the variational loss value associated with this VAE. Show all steps clearly!

B. Consider transforming a standard normal distribution in a flow model:

$$Pr(z) = \frac{1}{\sqrt{2\pi}} \exp\left[\frac{-z^2}{0.25}\right],$$

with the function:

$$x = f[z] = \frac{1}{1 + \exp[-2Z]}.$$

Find an expression for the transformed distribution $Pr(x)$ in terms of x.

[4+4= 8 Marks]

Q. **PART-A** Variational Autoencoder Loss calculation

Given, $\mu_z = [0, 0]$

covariance Matrix $= \Sigma_z = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

After training, $P_\phi(z/x) = P_\theta(x/z)$

It means, reconstruction loss is minimized.

& we need to calculate KL divergence part only.

$$\text{VAE Loss} = \underbrace{\text{Reconstruction Loss}}_{\downarrow \atop 0} + \text{KL Divergence}$$

$$\text{VAE Loss} = D_{KL}\left(\underbrace{N(\mu_z, \Sigma_z)}_{\text{Learned gaussian}} \,\|\, \underbrace{N(0,1)}_{\text{standard Normal}}\right)$$

Say, $q(z) = N(\mu, \Sigma)$

$p(z) = N(0,1)$

Hence $D_{KL}(q\|p) = \frac{1}{2}\left[Tr(\Sigma) + \mu^T\mu - k - \log(\det\Sigma)\right]$

$k=2$ (as latent space is 2D)

* $Tr(\Sigma) = 2+3 = 5$
* $\mu^T\mu = 0$ (since $\mu = 0$)
* $\log\det\Sigma = \log(2\cdot3) = \log(6)$

So, $D_{KL} = \frac{1}{2}[5+0-2-\log 6] = \frac{1}{2}[3-\log 6]$

$$D_{KL} = \frac{1}{2}(3 - 1.79)$$

$$= \frac{1}{2} \times 1.208 = 0.604$$

$$\boxed{\text{VAE Loss} = D_{KL} = 0.604}$$

PART-B Transformed Distribution in flow Model

$$P_r(z) = \frac{1}{\sqrt{2\pi}} e\left(\frac{-z^2}{0.25}\right)$$

$$\sigma^2 = 0.25 \Rightarrow \sigma = 0.5 \quad \& \quad \mu = 0$$

$$z \sim N(0, 0.25)$$

Transformation:

$$x = f(z) = \frac{1}{1 + \exp(-2z)}$$

$$P_r(x) = P_r(z) \cdot \left|\frac{dz}{dx}\right|$$

$$x = \frac{1}{1+e^{-2z}} \Rightarrow \frac{1}{x} = 1 + e^{-2z} \Rightarrow e^{-2z} = \frac{1-x}{x}$$

$$-2z = \log\left(\frac{1-x}{x}\right)$$

$$\Rightarrow \boxed{z = \frac{-1}{2} \log\left(\frac{1-x}{x}\right)}$$

Derivative $\left(\frac{dz}{dx}\right)$

$$\frac{dz}{dx} = \frac{d}{dx}\left(\frac{-1}{2}\log\left(\frac{1-x}{x}\right)\right)$$

using chain Rule

$$\frac{dz}{dx} = \frac{-1}{2}\cdot\frac{d}{dx}\left[\log(1-x) - \log(x)\right]$$

$$= \frac{-1}{2}\left[\frac{-1}{1-x} - \frac{1}{x}\right] = \frac{+1}{2}\left[\frac{1}{1-x} + \frac{1}{x}\right]$$

substitute z in terms of x into $P_r(z)$

$$z = \frac{-1}{2}\log\left(\frac{1-x}{x}\right)$$

$$P_r(z) = \frac{1}{\sqrt{2\pi\cdot 0.25}}\exp\left(\frac{-z^2}{0.25}\right)$$

$$= \frac{1}{\sqrt{0.5\pi}}\exp(-4z^2)$$

Substitute z:

$$z = \frac{-1}{2}\log\left(\frac{1-x}{x}\right) \Rightarrow z^2 = \frac{1}{4}\left[\log\left(\frac{1-x}{x}\right)\right]^2$$

$$P_r(z) = \frac{1}{\sqrt{0.5\pi}}\exp\left(-\log\left(\frac{1-x}{x}\right)\right)^2$$

$$Pr(x) = Pr(z) \cdot \left| \frac{dz}{dx} \right|$$

$$\boxed{Pr(x) = \frac{1}{\sqrt{0.5\pi}} \exp\left(-\left[\log\left(\frac{1-x}{x}\right)\right]^2\right) \times \frac{1}{2}\left(\frac{1}{x} + \frac{1}{1-x}\right)}$$